

Stellenbosch University

Department of Industrial Engineering

Non-linear Optimisation 414

Tutorial 4: 17 April 2026 (on §4.1–§4.3.1)

1. Use the golden section search heuristic for univariate functions to approximate the local maximum of the function $f(x) = -2x^3 + x + e^x$ within the unit interval $[0, 1.5]$. Terminate the search once the interval of uncertainty drops below $\varepsilon_{\text{tol}} = 0.05$ for the first time. Determine the number of calculations required before starting the heuristic. Write your output in tabular format similar to Table 4.1 in your textbook.

2. Maximise the function

$$f(x_1, x_2) = -x_1^3 + 3x_1 - x_2^2$$

by invoking a local search (Algorithm 4.3), starting out from the initial estimate $x^0 = [1 \ -0.2]^T$ specified. Discretise the search space according to an interval lattice structure in which each subinterval has length $\varepsilon = 0.3$. Half the resolution of the lattice structure if no improving iteration is found and stop as soon as the resolution of the lattice structure drops below 0.1. Write your output in tabular format similar to Table 4.4 in your textbook.

3. Minimise the function

$$f(x_1, x_2) = x_1^2 + x_2^2 - 2x_1 + 4x_2 + 1$$

by invoking a tabu search (Algorithm 4.4), starting out from the initial estimate $x^0 = [0.8 \ -1.5]^T$. Discretise the search space according to an interval lattice structure in which each subinterval has length $\varepsilon = 0.3$. Take the tabu tenure of the reversal of a search direction as 2 and implement an aspiration criterion of always accepting a move if it improves the incumbent solution. Work accurately to three decimal places and terminate the search once the incumbent could not be improved over two consecutive iterations. It is not necessary to increase the resolution of the lattice structure at any point. Produce a table of algorithmic output similar in structure to Table 4.5.

4. Given the population $\begin{bmatrix} 0.3 \\ -0.2 \end{bmatrix}$, $\begin{bmatrix} -0.9 \\ 0.5 \end{bmatrix}$, $\begin{bmatrix} -0.4 \\ 0.5 \end{bmatrix}$, $\begin{bmatrix} 0.9 \\ 0.7 \end{bmatrix}$, and $\begin{bmatrix} -0.4 \\ -0.5 \end{bmatrix}$, and the parent solutions

$$\mathbf{x}^1 = \begin{bmatrix} -0.4 \\ -0.5 \end{bmatrix} \text{ and } \mathbf{x}^2 = \begin{bmatrix} -0.9 \\ 0.5 \end{bmatrix},$$

create 2 offspring solutions $\mathbf{y}^1$ and $\mathbf{y}^2$ by performing intermediate crossover with parameters $\alpha^1 = 0.25$ and $\alpha^2 = 0.6$.